

Title :**StockSage – An Agentic AI Assistant for Stock Market Analysis****Abstract :**

In the rapidly evolving financial ecosystem, individual investors and analysts often face challenges in processing vast amounts of data to make informed decisions. StockSage addresses this gap by introducing an Agentic AI system that autonomously performs stock market research, sentiment analysis, and investment strategy generation with minimal user input.

At its core, StockSage is powered by a multi-agent architecture that integrates large language models (LLMs) with real-time financial data sources. Upon receiving a stock ticker or company name, the system's Stock Analyzer Agent retrieves fundamental metrics, valuation ratios, and historical trends, presenting them through intuitive visualizations. Simultaneously, the Sentiment Analyzer Agent gathers and classifies public sentiment from financial news and social media using natural language processing techniques. Finally, the Strategy Generator Agent evaluates the user's risk profile and investment horizon to deliver personalized insights, including buy/sell suggestions, target prices, and asset allocation strategies.

To enhance usability, StockSage features an accessible interface for interactive input and report delivery. An Auto-Report Generator compiles all findings into a structured, downloadable format suitable for presentations and recordkeeping.

StockSage showcases the transformative potential of autonomous AI agents in democratizing financial intelligence and enabling users to make informed, data-driven investment decisions with ease.

Keywords :

Stock Market Analysis, Agentic AI System, Sentiment Analysis, Financial Data Visualization, Investment Strategy Generator, Large Language Models (LLMs), yfinance / Alpha Vantage , GPT-4 / OpenAI.

Key Features :

1. Stock Analyzer Agent

- Automatically fetches company profile, financial metrics, valuation ratios, and historical trends based on ticker symbol (e.g., TSLA, TCS).
- Uses APIs like yfinance or Alpha Vantage.

2. Sentiment Analyzer

- Scrapes news headlines and social media.
- Uses NLP to classify sentiment as Positive / Negative / Neutral.
- Visualizes sentiment over time using graphs.

3. Investment Strategy Generator

- Uses OpenAI GPT to provide personalized investment advice.
- Suggests buy/sell decisions, target price, and reasoning.
- Considers user risk profile and time horizon.

4. Auto-Report Generator

- Compiles all analysis and AI-generated strategies.
- Exports results as a clean, downloadable PDF report.

5. User-Friendly Interface (via Streamlit)

- Interactive dashboard for input, insights, and report viewing.
- Minimal user input required; results are generated automatically.

Workflow :

